



Intel's systems. OpenAI's Triton worked primarily with Nvidia, so we built much of the required infrastructure ourselves—particularly for CPUs—by creating computing, communication, and networking libraries from scratch. Even with this foundation, we needed to ensure AI models run efficiently in real-world situations, so we developed a simulation system to help optimise performance for internal development and end-user deployments.”

The startup is building a platform to minimise hardware dependency for GenAI infrastructure—a challenge that remains difficult to overcome with existing systems. Systems built on Nvidia CUDA are locked into Nvidia hardware, and even deploying across different Nvidia chips can be challenging due to varying CUDA versions. Mixing hardware from other vendors like Nvidia, AMD, and Intel—or combining GPUs, HPUs, and CPUs—is currently unfeasible.

The team claims to be the only company addressing this challenge through heterogeneous



Jithin V.G., Founder & CEO, Bud Ecosystem

parallelism and clustering, along with SLO-based management that abstracts hardware details. This approach means users no longer need to worry about the underlying hardware. It is similar to how operating systems once addressed mainframe limitations by offering hardware abstraction and portability. Today, GenAI faces a comparable issue: models developed for Nvidia platforms often cannot be deployed on alternative systems. The startup aims to address this by

enabling AI applications to operate seamlessly across diverse hardware environments.

Open-sourcing GenAI presents significant challenges. Although the startup required numerous contributors, it faced difficulty sourcing them. In India, while many use open source tools, few engage in building large-scale systems. For Bud Ecosystem, the challenge was both technical and cultural—building complex systems while nurturing a stronger open source community in India.

Bud Ecosystem focuses on core AI research and model architectures tailored for India and similar markets. Their work targets models that consume less energy, require less computational power, and can run on edge devices. Rather than following the large data centre approach, they believe India can leverage billions of devices as a decentralised computing network. They believe innovation should precede growth, which will follow naturally when meaningful problems are addressed.

—Nidhi Agarwal

3 XBATTERY'S INDIGENOUS BMS THAT'S CHEMICAL AGNOSTIC?

Hyderabad-based deep-tech startup Xbattery is developing intelligent, secure BMS solutions to power India's clean energy and electric mobility shift.

Founded in August 2024 by Satish Reddy and Sonu Mishra, Xbattery Energy Pvt Ltd is an emerging deep-tech firm focused on designing indigenous battery management systems (BMS) for electric vehicles and stationary energy storage.

The company's flagship product, BharatBMS, is a high-tech printed circuit board (PCB) entirely designed and manufactured

in-house. It functions as the central electronic control unit within battery packs, ensuring intelligent and efficient battery management. BharatBMS is available in the market, priced between ₹2000 and ₹25,000, depending on the configuration. The cost varies based on voltage requirements—lower for scooters and higher for trucks or larger vehicles.

Another key offering from Xbattery is a 5kWh home energy

storage system, developed as a modern replacement for traditional inverter-based backup systems. This lithium-based system incorporates the BharatBMS and is assembled in-house using imported battery cells. It is expected to launch in the market within the next three months.

Satish highlights BharatBMS's configurability and broad compatibility with various voltages and